

Unit 1: Introduction to Machine Learning

Detailed Notes

1. What is Machine Learning?

Machine Learning (ML) is the field of study that gives computers the ability to learn from data without being explicitly programmed.

2. Types of Machine Learning:

- Supervised Learning: Learns from labeled data.
- Unsupervised Learning: Finds patterns in unlabeled data.
- Reinforcement Learning: Learns from interactions using rewards.

3. Applications:

- Spam filtering, handwriting recognition, fraud detection, recommendations, medical diagnosis, stock prediction.

4. Challenges:

- Overfitting: Performs well on training but poorly on test data.
- Underfitting: Too simple to learn.
- Bias-Variance Trade-off: High bias -> underfitting; High variance -> overfitting.

5. Concept Learning:

- Task: Inferring a boolean function.
- Hypothesis Space (H): All possible hypotheses.

6. Version Space:

- Set of hypotheses consistent with all observed training examples.
- General Boundary (G) and Specific Boundary (S).

7. Inductive Bias:

- Assumptions a learner makes to generalize beyond data.

8. Generalization:

- Ability of model to perform well on unseen data.

Flashcards

Q: What are the three main types of ML?

A: Supervised, Unsupervised, Reinforcement.

Q: What causes overfitting?

A: A model that's too complex and fits noise in training data.

Q: What is hypothesis space?

A: The set of all possible models a learner can select from.

Q: What is the version space?

A: Hypotheses consistent with all training examples.

Q: What is inductive bias?

A: Assumptions a learner makes to generalize beyond data.

Practice Questions

A. Short Answer:

1. Define machine learning and its types.
2. What is the hypothesis space in concept learning?
3. Explain version space with an example.
4. Describe inductive bias.

B. Multiple Choice:

1. Which is supervised learning?

a) Clustering

b) Association Rules

c) Linear Regression [Correct]

d) PCA

2. Which issue occurs when a model is too complex?

- a) Underfitting
- b) Overfitting [Correct]
- c) No fitting
- d) Generalization

C. True or False:

- Inductive bias helps the learner to generalize. [Correct]
- Overfitting happens when a model is too simple. [Incorrect]

Python Example: Find-S Algorithm

```
def find_s(examples):
    hypothesis = ['0'] * len(examples[0][0])
    for x, y in examples:
        if y == 'Yes':
            for i in range(len(x)):
                if hypothesis[i] == '0':
                    hypothesis[i] = x[i]
                elif hypothesis[i] != x[i]:
                    hypothesis[i] = '?'
    return hypothesis

data = [
    (['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same'], 'Yes'),
    (['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same'], 'Yes'),
    (['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change'], 'No'),
    (['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change'], 'Yes')
]

print("Final Hypothesis:", find_s(data))
```